

Multi-spacecraft study of CME-CME interactions, ICME-in-sheath and associated GLE #77 during the 11-12th Nov 2025 event

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Date	Flare	CME Speed	Role
9 Nov - 7:36 UT	X1.7	804 km/s	CME ₁
10 Nov - 9:48 UT	X1.2	1320 km/s	CME ₂
11 Nov - 10:04 UT	X5.1	1845 km/s	CME ₃

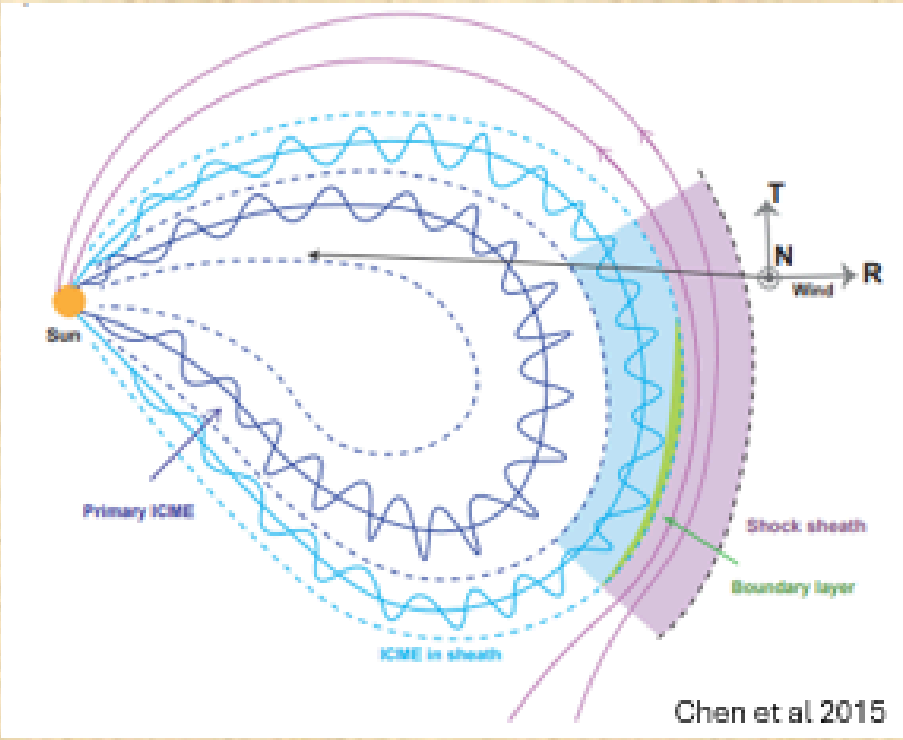
Complex
Ejecta (CE)

ICME signatures:

- High jump in Plasma speed
- High B fluctuations
- Higher Density of Particles
- Low plasma beta ratio
- Higher Ion Flux

ICME-in-sheath signatures:

- Short in duration
- ICME signatures are often lost
- SW parameters enhancements



CE interacted with CME₃ during 12 Nov during on its way to Earth

Geospace's Response

First Shock : ~23:48 UT, 11 Nov

Dst_{min}: ~217 nT

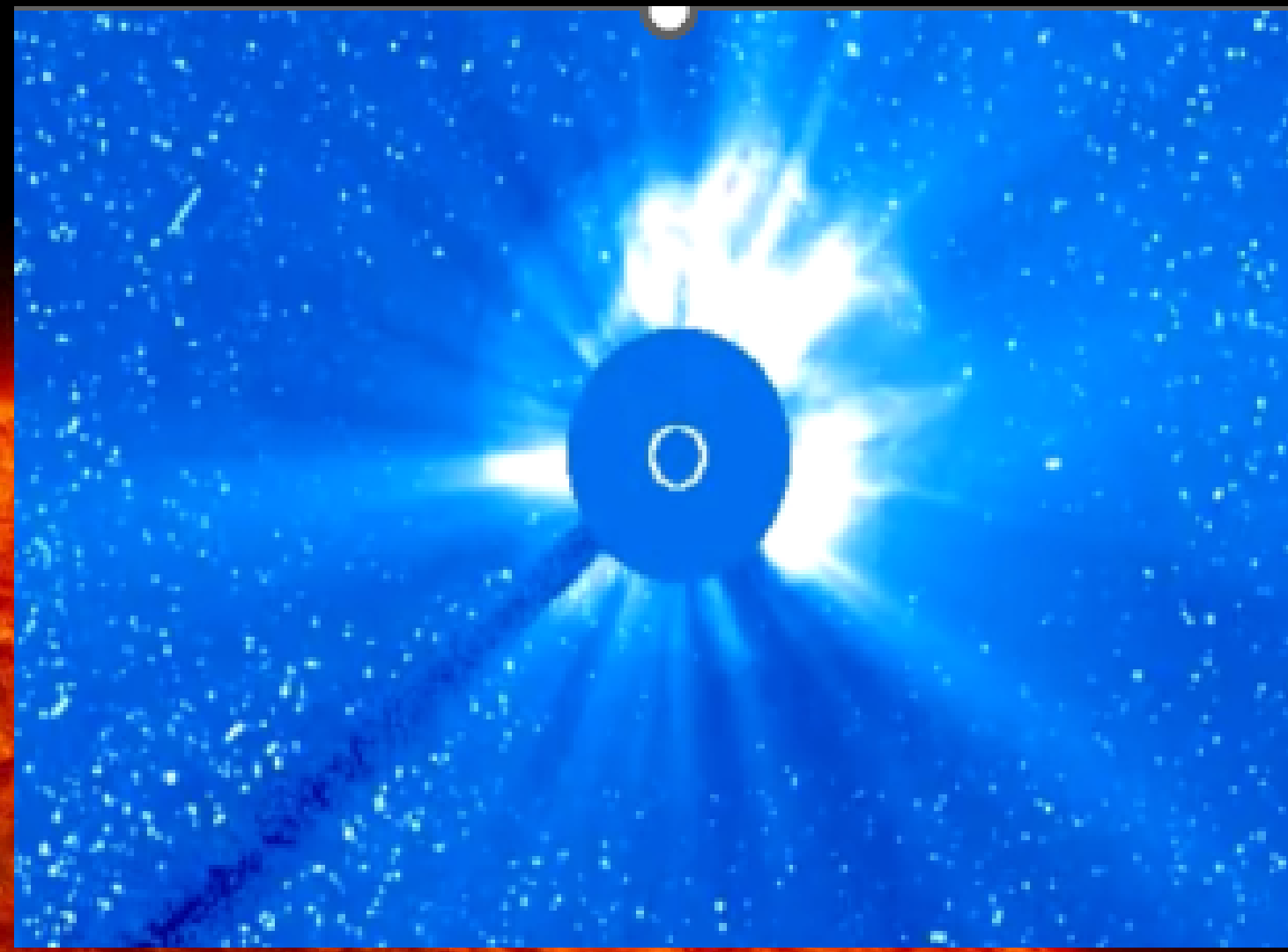
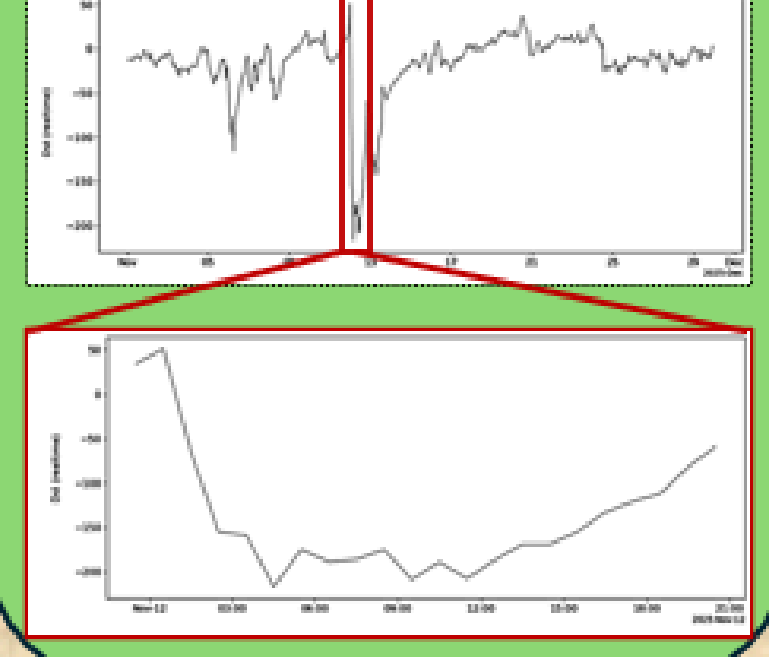


Fig 1: LASCO coronagraph observation showing the outward propagation of the CME from the solar corona. The coronagraph view provides the remote-sensing evidence for the eruption and large-scale expansion of the CME prior to its subsequent interplanetary propagation.

Mission's Data used:

*Aditya-L1 (MAG, ASPEX) *WIND (SWE) *OMNI

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2025-11-12 13:00Z

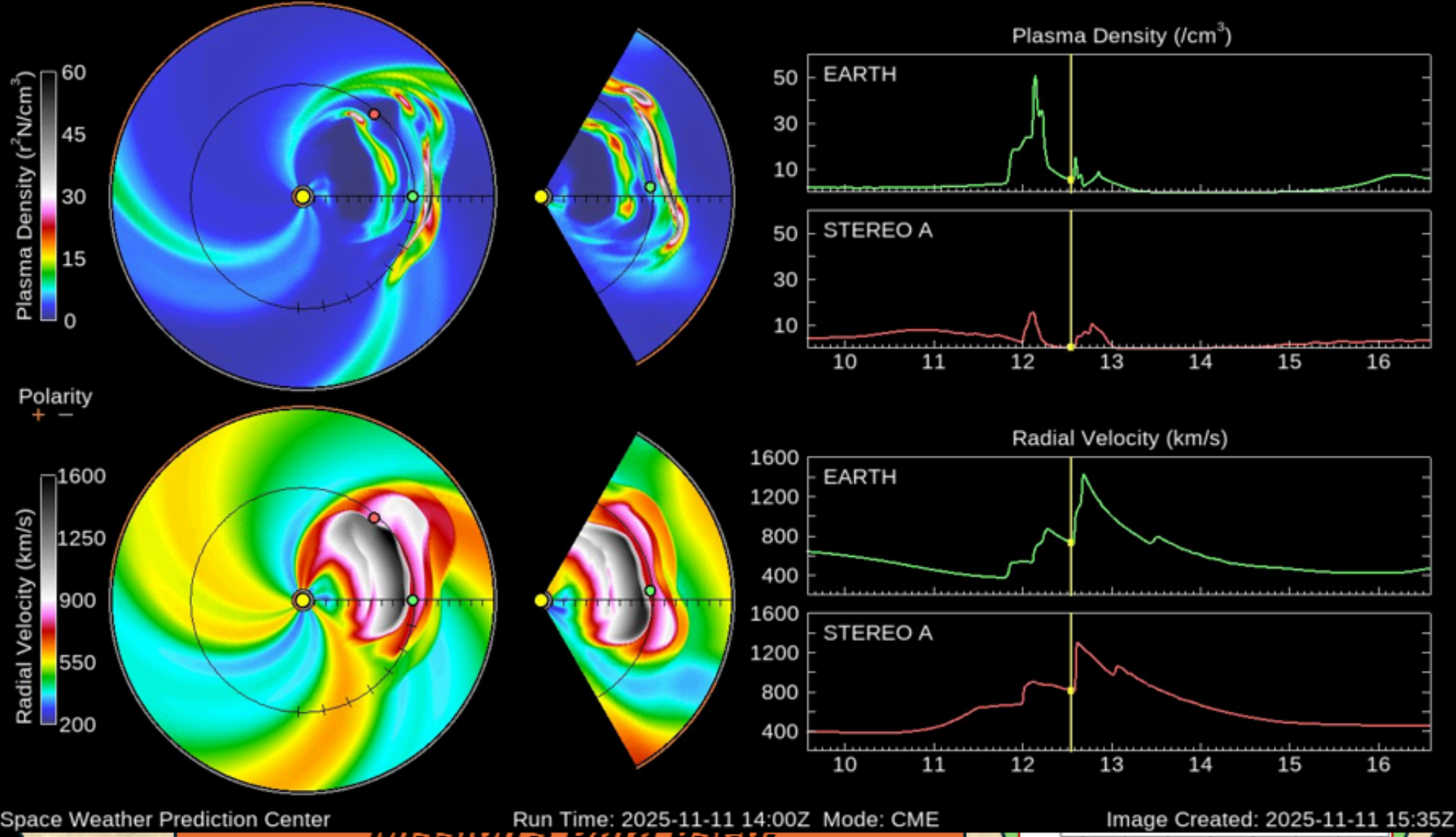


Fig 2: WSA-ENLIL simulation illustrating the large-scale propagation of the successive CMEs and the development of interacting/merged structures. The simulation supports the formation of a preconditioned heliosphere before the arrival of the 11 November CME and provides global context for the complex structures identified in the multi-spacecraft observations.

ADITYA-L1

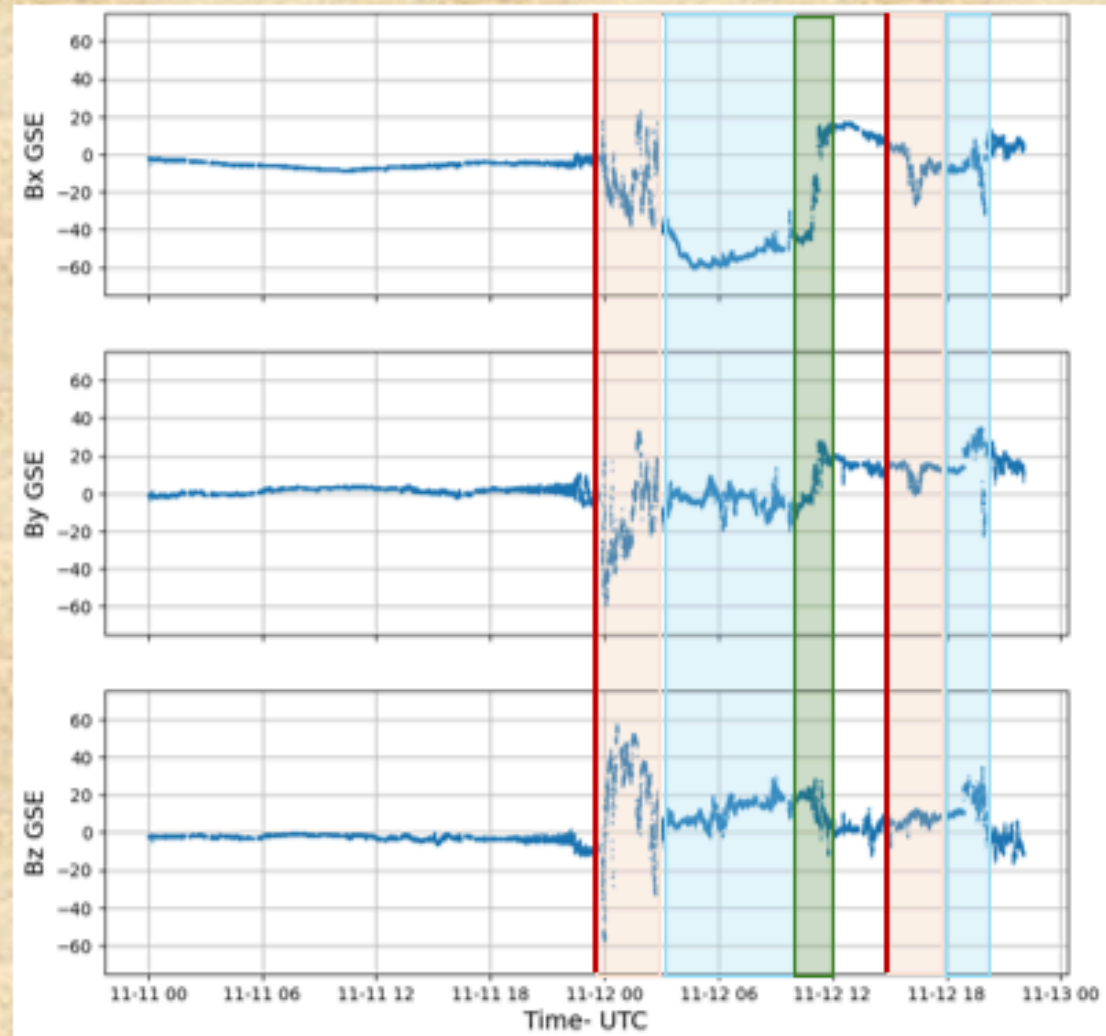


Fig. 1: (a) B_x , B_y , and B_z components of the interplanetary magnetic field (IMF) measured by the MAG instrument.

OMNI

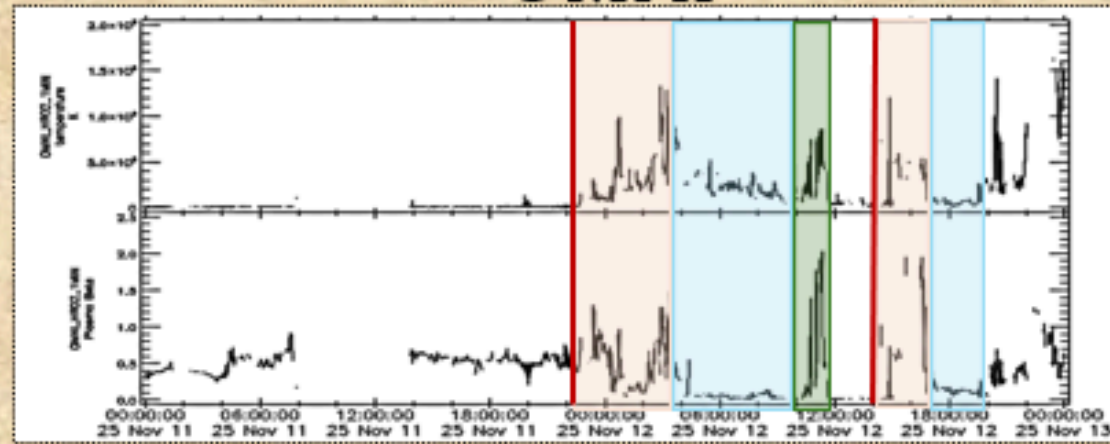


Fig. 3: (a) Plasma temperature and (b) plasma beta (β) observed from the OMNIWeb dataset.

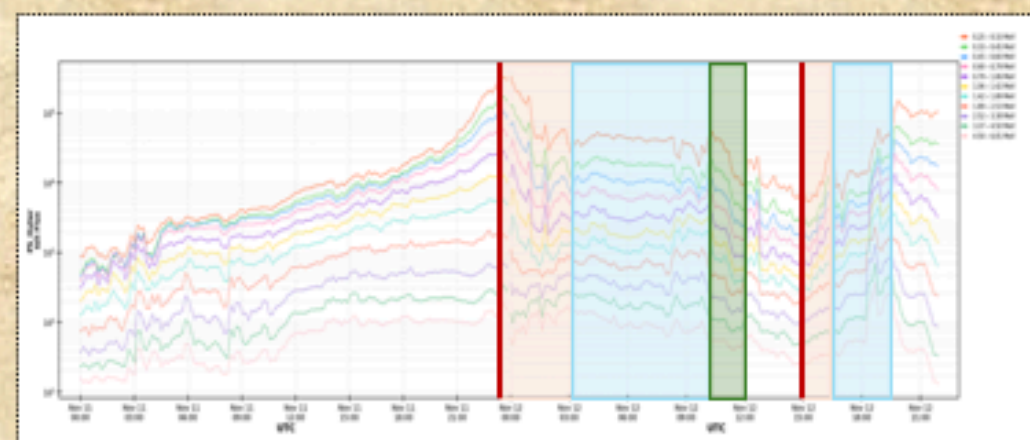


Fig. 2: (b) energy-resolved ion flux measurements from the STEPS instrument along the Parker spiral (PS) direction.

WIND

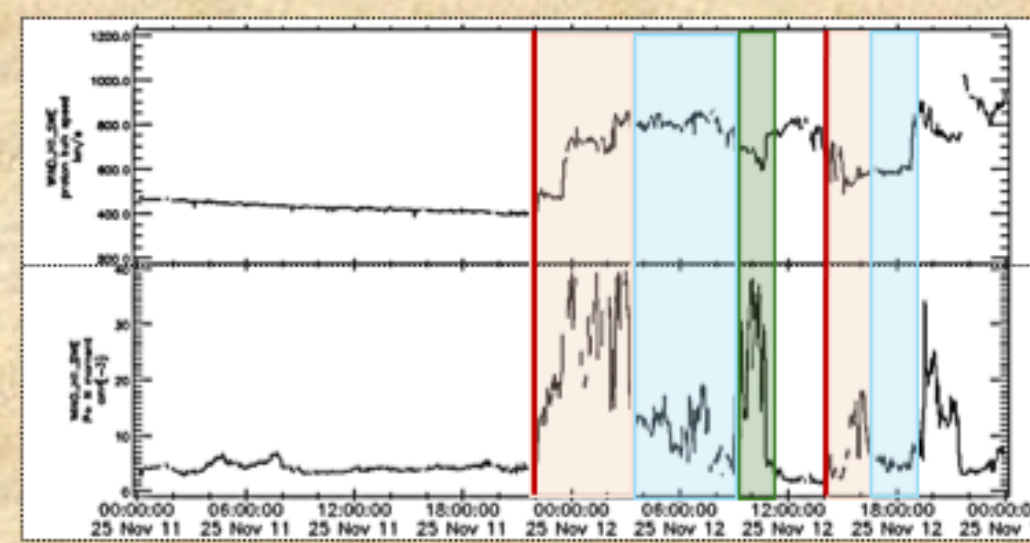
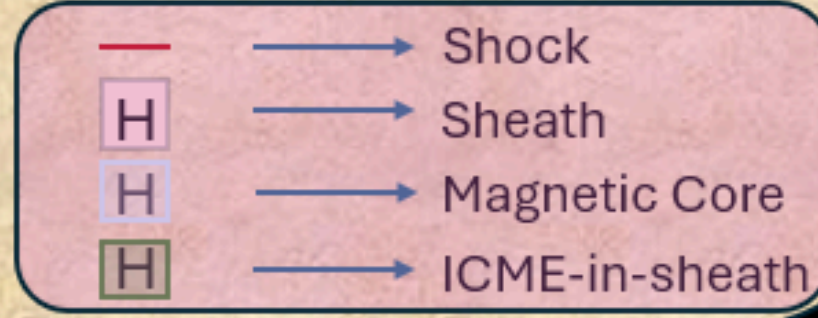


Fig. 4: (a) Proton bulk speed (km s^{-1}) and (b) proton number density, N_p (cm^{-3}), observed by the Wind-SWE



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WHY IS GLE #77 UNUSUAL?

- GLE #77 -> Rare double-peak profile observed at SOPB, and SOPO NMs
- DOMB & DOMC NMs exhibit a delayed response without double-peak structures.

Impulsive phase

- Rapid and Highly anisotropic
- Indicates a localized acceleration source.
- Particles escape down a single, well-connected Parker Spiral magnetic field line.

Gradual phase

- Broader and more widespread
- Consistent with progressive broadening of particle access/distribution.
- Indicates acceleration source is no longer a single point on the Sun, but a broad, moving shock wave

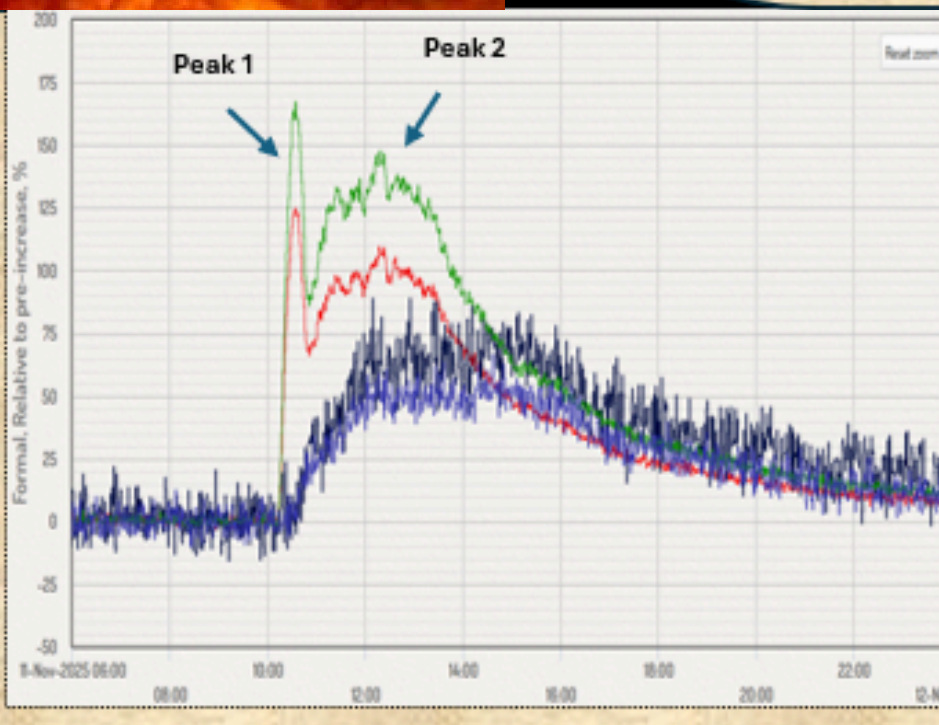


Fig 5: Multi-Channel Neutron Monitor Response to GLE #77

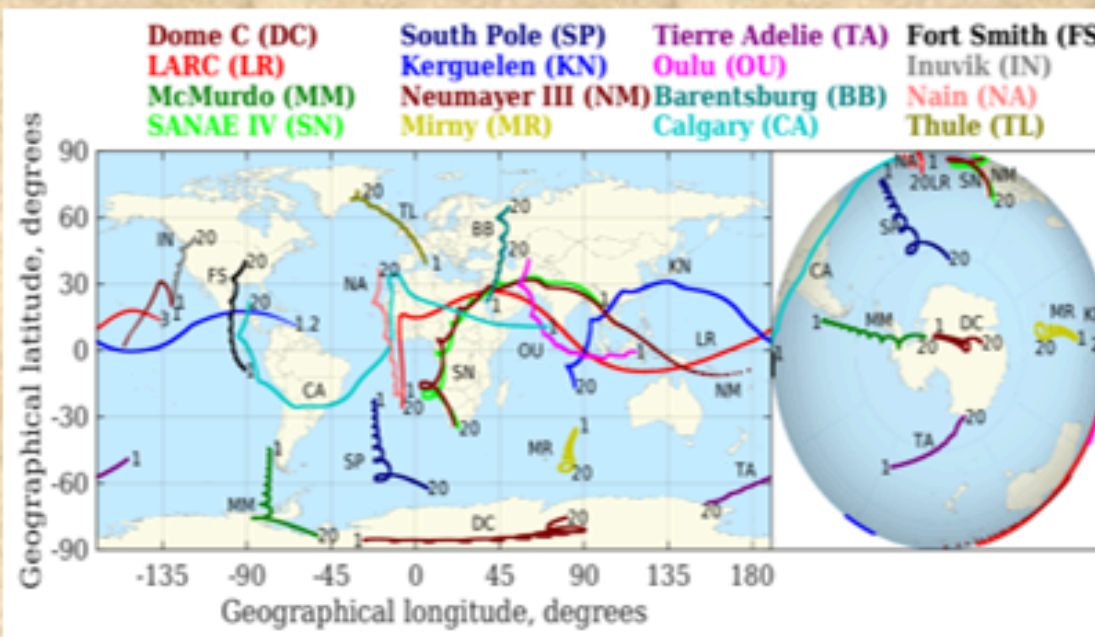


Fig 6: Asymptotic acceptance curves for several cosmic ray stations

In simple nominal Parker-spiral picture, direct magnetic access from N23W24 is less favourable.

This indicates that pre-existing disturbed environment the accelerate the particle transport of GLE-77 particles toward Earth

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Solar Source Locations of Major Ground-Level Enhancement Events

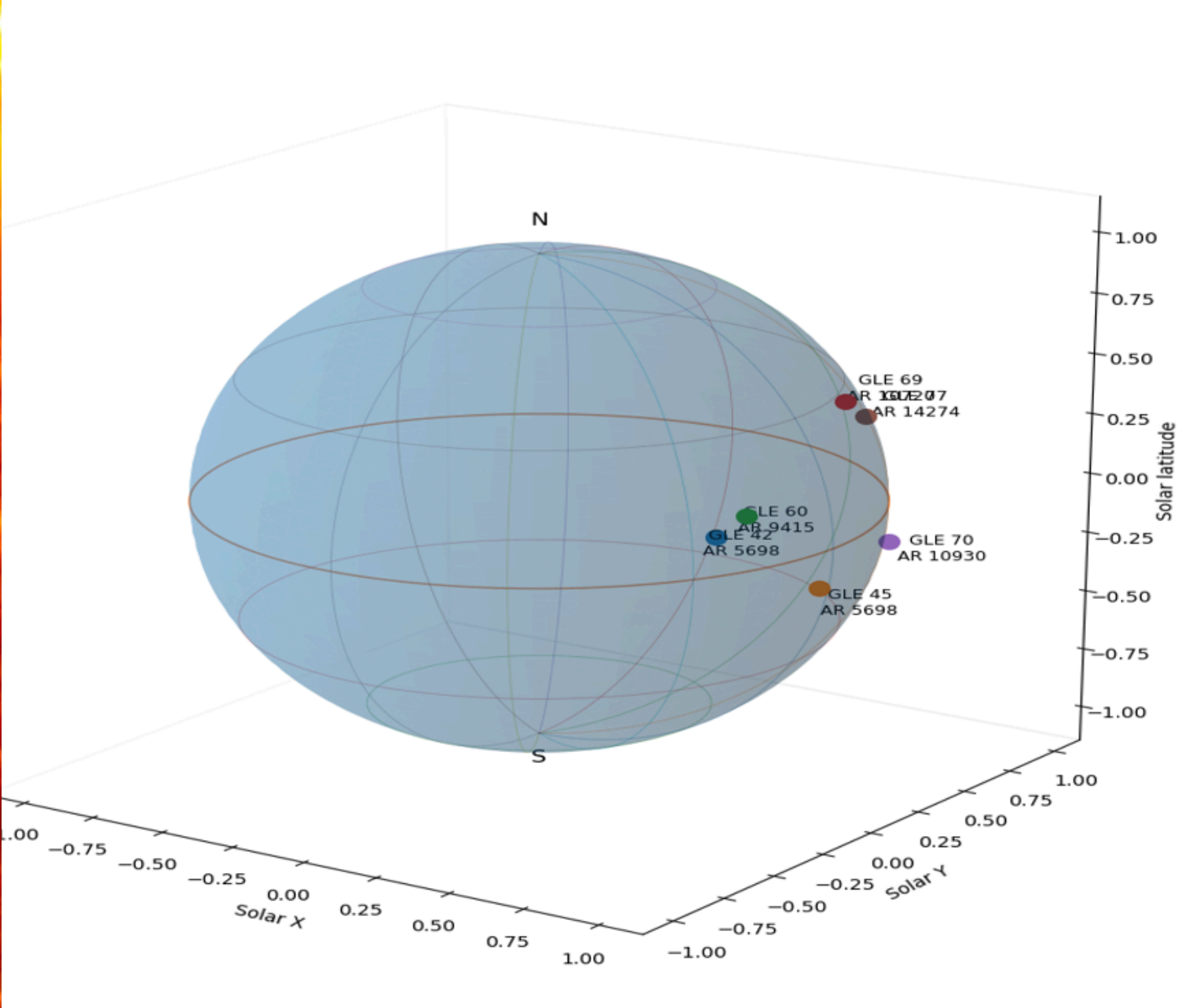


Fig 3: Comparison of heliographic source locations for major GLEs over the past two decades. GLE #77 originated from N23W24, noticeably eastward of several previous major GLE sources located toward the western solar limb. This unusual source geometry, together with the complex pre-existing heliospheric conditions, provides motivation for investigating the observed GLE #77 particle transport and anisotropy.